

Yanna Ding

Portfolio: <https://dingyanna.github.io>

Work Authorization: U.S. Permanent Resident

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Theory to production: learning systems, LLM security, AI agents.

Education

- ▷ Rensselaer Polytechnic Institute (RPI) Spring 2022 - Current
Ph.D. program in Computer Science GPA: 4.0/4.0
- ▷ University of Toronto (UofT) 2017 - 2021
Honours Bachelor of Science in Computer Science and Mathematics GPA: 3.92/4.0

Experience

- ▷ AI SECURITY & LLM AGENTS Samsung Research America, Research Engineer Intern (Summer 2026)
 - Designed a **lightweight keyword pre-filter** for prompt injection detection under **extreme class imbalance**; implemented **cascade two-stage architecture** to improve precision while maintaining end-to-end recall accountability
 - Extended a privileged **Android AI agent** with new context-driven capabilities, implementing gesture-triggered overlays, screen context extraction, and automated task workflows (e.g., form filling, morning brief) within a **multi-module LLM orchestration architecture**
 - ★ *Skills: Prompt Injection Defense, Cascade Classifiers, Class Imbalance, Android AI Agents, On-device LLM*
- ▷ PROMPT INJECTION DETECTION & LLM AGENT EVALUATION Samsung Research America, Research Engineer Intern (Spring 2026)
 - Investigated **prompt injection attacks** in LLMs using both public QA/RAG benchmarks (e.g., deepset, Qualifire) and internally curated adversarial datasets, evaluating model robustness under malicious input perturbations
 - Built a **scalable benchmarking engine** to support **hyperparameter sweeps**, leveraging **Ray Tune** to scale experimentation across configurations
 - Developed an **automated evaluation pipeline** for LLM-based agents, enabling **log-based grading** and **structured analysis of agent behavior**
 - ★ *Skills: LLM Benchmarking, Prompt Injection Defense, Agent Evaluation Systems, Scalable Experimentation*
- ▷ MOBILE SECURITY Samsung Research America, Research Engineer Intern (Fall 2025)
 - Rapidly prototyped **end-to-end ML pipeline** for Android mobile threat detection (**preprocessing, model training, on-device deployment**), reducing False Positive Rate by **2%** through **data cleaning** and **threshold tuning**
 - Collaborated with security researchers and engineers to evaluate **Graph Neural Network (GNN)** architectures on **13K+ app dataset**; delivered team's **reference workflow**
 - ★ *Skills: ML Pipelines, Graph Neural Networks, On-device ML, Mobile Threat Detection*
- ▷ IN-CONTEXT LEARNING & LLM REASONING OPTIMIZATION IBM Research, Research Extern (Summer 2024, 2025)
 - Developed a **nested multilevel optimization framework** for chain-of-thought learning, reducing out-of-distribution errors by **40%** on structured reasoning tasks
 - Analyzed Transformer in-context learning for sequential data, characterizing architectural limits and proving the **NP-hardness** of parameter recovery in **Markovian** dynamical functions under linear self-attention
 - ★ *Publication: NeurIPS'25, ICASSP'25*
 - ★ *Skills: Optimization for Sequential Models, In-Context Learning, Learning Theory*
- ▷ ARCHITECTURE-AWARE LEARNING CURVE EXTRAPOLATION UCLA, Visiting Student (Spring 2024)
 - Designed a **Graph Neural ODE** model to predict training learning curves from partial runs, enabling early architecture selection
 - Achieved up to **20× reduction in search cost** with **59.63% improvement in extrapolation accuracy** over baselines
 - ★ *Publication: AAAI'25*
 - ★ *Skills: Graph Neural Networks, Neural ODEs, Learning Curve Extrapolation*
- ▷ DECODING-FREE CANDIDATE SELECTION FOR GENERATIVE RETRIEVAL UCLA, Visiting Student (Spring 2024)
 - Developed **logits-based candidate selection** methods that bypass autoregressive decoding for generative retrieval
 - Achieved **+29.25 point recall improvement** with up to **145× inference speedup** across QA and clinical datasets, scaling to 94k candidates
 - ★ *Publication: ACL'25*

★ *Skills: LLM Inference Optimization, Generative Retrieval, Candidate Selection*

- ▷ **PARAMETER INFERENCE IN NETWORKED DYNAMICAL SYSTEMS** **RPI**, Research Assistant (Spring 2022 - Present)
 - Developed **surrogate objective functions** to infer dynamical parameters from noisy steady-state observations in large-scale networks
 - Achieved up to **300× speedup** over full ODE-based optimization while preserving steady-state prediction accuracy
- ★ *Publication: Physics Review E'24, ASONAM'23*
- ★ *Skills: Dynamical Systems, Parameter Inference, Surrogate Modeling, Optimization*

Selected Publications

- ▷ Optimality and NP-Hardness of Transformers in Learning Markovian Dynamical Functions
Ding Y., Lu S., Lu Y., Nowicki T., Gao J. NeurIPS'25 (acceptance rate ~24.52%)
- ▷ Inferring from Logits: Exploring Best Practices for Decoding-Free Generative Candidate Selection
Ma M.*, Ding Y.*, Huang Z., Gao J., Sun Y., Wang W. ACL'25 (acceptance rate ~22%)
* (Equal contribution)
- ▷ Epigraph Based Multilevel Optimization for Enhancing Chain-of-Thought Reasoning
Lu S., Ding Y., Horesh L., Gao J., Magdon-Ismail M. ICASSP'25 (acceptance rate ~48%)
- ▷ Architecture-Aware Learning Curve Extrapolation via Graph Ordinary Differential Equation
Ding Y., Huang Z., Shou X., Guo Y., Sun Y., Gao J. AAAI'25 (acceptance rate 23.4%)
- ▷ Efficient Parameter Inference in Networked Dynamical Systems via Steady States
Ding Y., Gao J., Magdon-Ismail M. Phys. Rev. E'24 (acceptance rate 20-30%)
- ▷ Learning Network Dynamics via Noisy Steady States
Ding Y., Gao J., Magdon-Ismail M. ASONAM'23 (acceptance rate 36.5%)

Honours and Awards

- ▷ Selected as a graduation spotlight student UofT, Spring 2021
- ▷ Dean's List Scholar, Faculty of Arts and Science UofT, 2018 - 2021
- ▷ Mitacs Research Training Award UofT, Fall 2020
- ▷ Department of Computer Science, Undergraduate Research Award UofT, Summer 2020
- ▷ The Chancellor's Scholarship for high academic achievement St. Hilda's Fund, 2019-2020
- ▷ Admission Scholarships UofT, Fall 2017

Service

- ▷ Reviewer for ICML 2022, KDD 2024, COLM 2025

Skills

- ▷ **LLMs & NLP:** In-context Learning, Chain-of-Thought, Sequence Modeling, Efficient Inference
- ▷ **AI Security:** Prompt Injection Defense, Cascade Classifiers, Class Imbalance, LLM Robustness Evaluation
- ▷ **AI Agents:** LLM Orchestration, Agent Evaluation, Tool Dispatch, On-device LLM
- ▷ **ML Foundations:** Optimization, Computational Complexity (NP-hardness), Markov Chains, Data Cleaning, Feature Engineering, Model Training and Evaluation
- ▷ **Tools:** PyTorch, TensorFlow, HuggingFace, PyTorch Geometric, Scikit-learn, Weights & Biases, TensorFlow Lite
- ▷ **Programming:** Python, L^AT_EX, MATLAB, Kotlin